

OLYMPIA OYSTER · LOW-COVERAGE WHOLE-GENOME SEQUENCING

Assessing environmental drivers of Olympia oyster population genetic diversity in Puget Sound, WA

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THE QUESTION

Where should restoration seed come from?

Match the environment

Source seed from beds with similar temperature, salinity and food, on the idea that oysters are tuned to local water.

Works if water conditions shape oyster genetics.

Match the genetic group

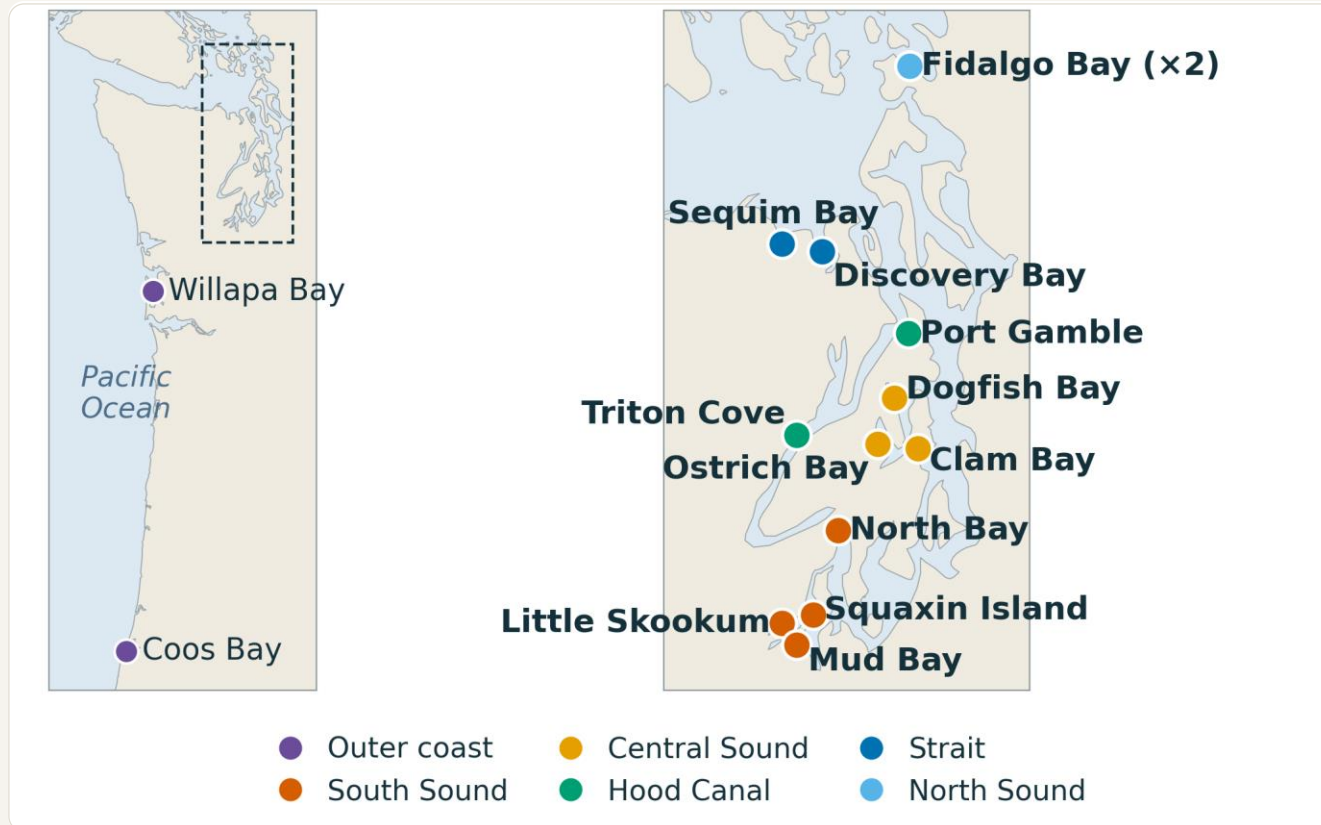
Source seed from the same genetic “family” of populations, keeping natural groupings intact.

Works if shared history shapes oyster genetics.

We asked which of these better explains the genetic differences we see among Puget Sound oyster beds.

SAMPLING DESIGN

15 collections span the Salish Sea and outer coast



109

oysters analysed (one low-depth sample dropped)

15

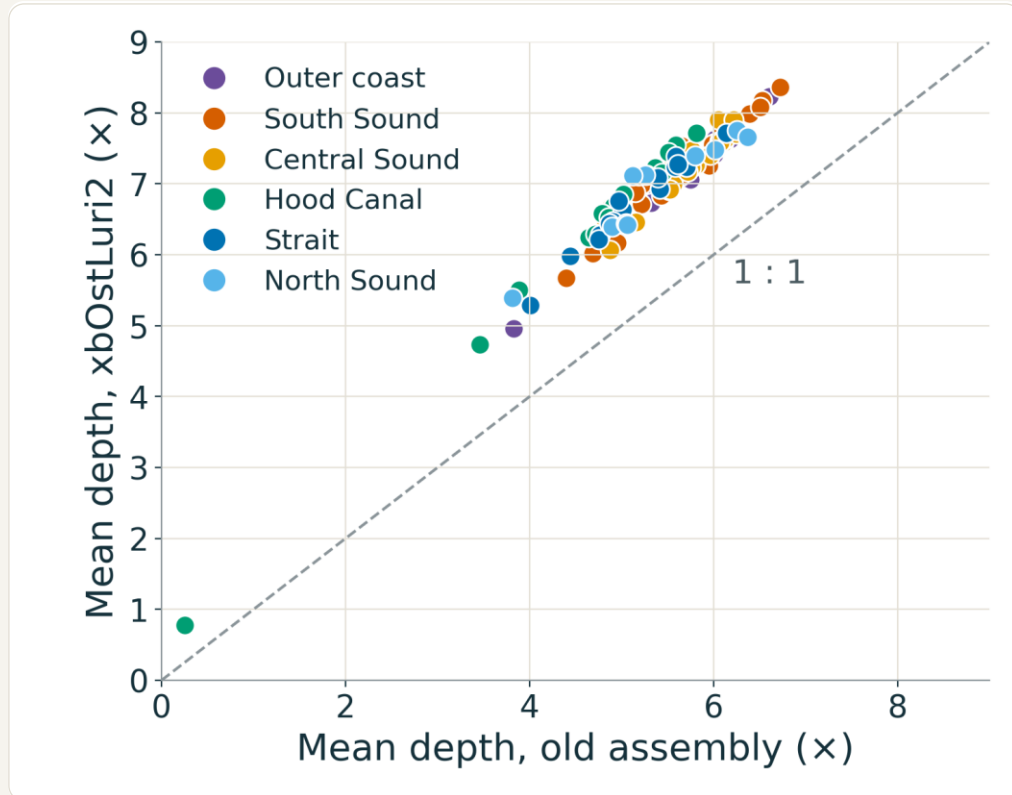
collections: 13 in the Salish Sea plus Willapa Bay and Coos Bay

~7x

mean sequencing depth per oyster

NEW REFERENCE GENOME

A chromosome-level genome raises depth and resolution



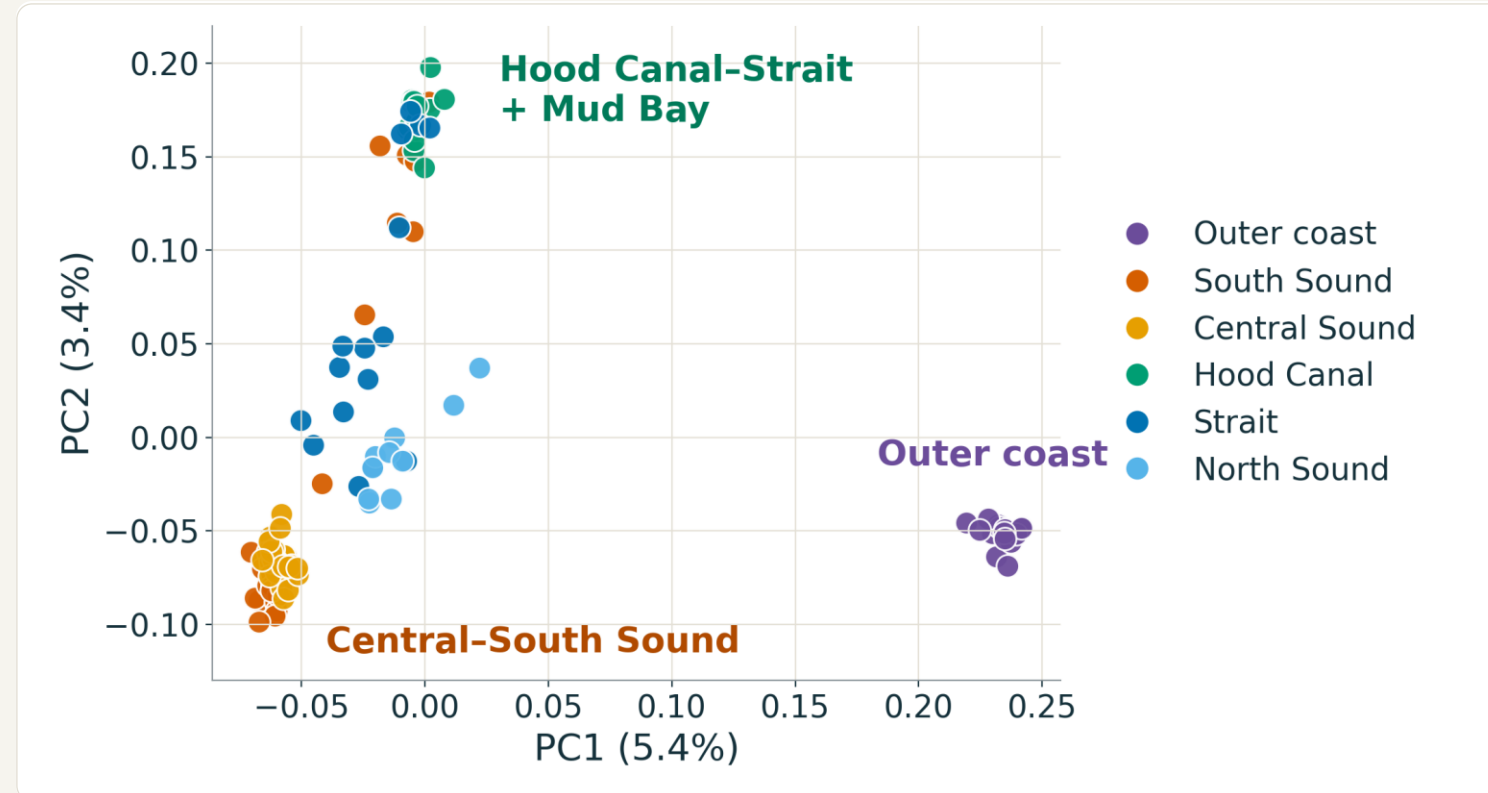
Metric	Olurida_v081	xbOstLuri2
Assembly	contigs, N50 12.9 kb	10 chromosomes, 995 Mb
Mean depth	5.4×	7.0×
Reads mapped	—	98.6%

Every oyster gained depth ($\approx 1.3\times$ on average), and SNPs can now be placed by chromosome and position.

Reference: GCA_061535525.1 (USDA-ARS / PSRF)

POPULATION STRUCTURE

Genomes split the outer coast from the Salish Sea



PC1 • 5.4%

Willapa Bay and Coos Bay separate from every Salish Sea site.

PC2 • 3.4%

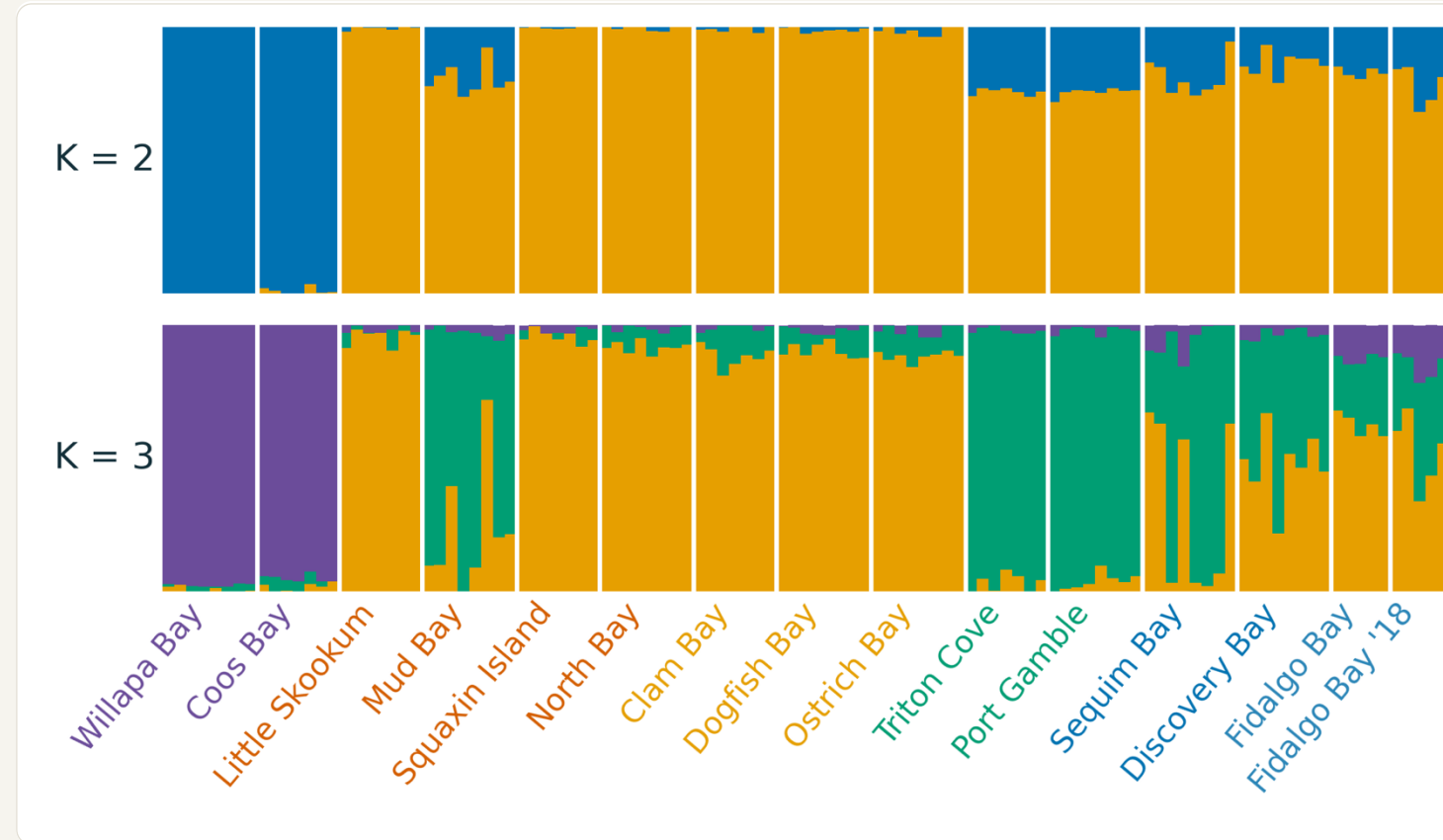
Hood Canal, Sequim, Port Gamble and Mud Bay split from Central and South Sound.

In between

Fidalgo Bay and Discovery Bay are intermediate.

POPULATION STRUCTURE

Admixture resolves three groups; Mud Bay joins Hood Canal



K = 2

Outer coast versus the Salish Sea.

K = 3

Orange: Central–South Sound.

Green: Hood Canal, Sequim, Mud Bay.
Discovery and Fidalgo mix both.

A puzzle

Mud Bay lies in South Sound yet groups with Hood Canal. Is this a legacy of past transfers?

FINDING 1

Oysters fall into three genetic groups

Outer coast	Central & South Sound	Hood Canal & Strait	Mixed
Willapa Bay Coos Bay (OR) <i>Separate stock from Puget Sound</i>	Clam Bay area Dogfish Bay Ostrich Bay North Bay Squaxin Island Little Skookum <i>Tight, closely related group</i>	Triton Cove Port Gamble Bay Sequim Bay Mud Bay* <i>*Mud Bay sits in South Sound</i>	Fidalgo Bay Discovery Bay <i>Share ancestry with both Sound groups</i>

FINDING 2

The outer coast is a different stock

3–4×

more different

Coast oysters differ from Puget Sound oysters 3–4 times more than Puget Sound beds differ from one another.

≈ same

Willapa Bay and Coos Bay

The two coastal sites, in different states, are as similar as neighbouring Puget Sound bays.

2 groups

inside the Sound

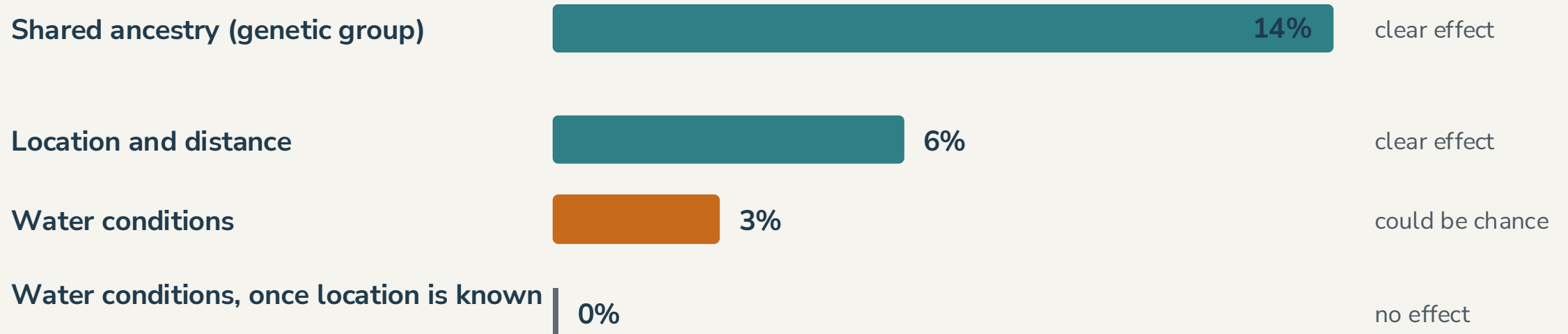
Differences within Puget Sound are real but modest, mostly between the two groups on the last slide.

Genetic differentiation (weighted F_{st}): coast vs. Puget Sound 0.11–0.14; among Puget Sound sites 0.034–0.088; Willapa vs. Coos Bay 0.037.

FINDING 3

History and geography outweigh water conditions

Share of the genetic differences between sites that each factor explains



Water conditions = summer temperature, salinity swings and summer food (chlorophyll), 2015–2018. Redundancy analysis on 14 sites. The result holds after accounting for collection year (2018 vs. 2024).

FINDING 4

No clear “local adaptation” genes yet

0

standout markers

Of 7.3 million markers scanned, none stood out as tied to water conditions.

Weak

salinity hint

Sites with bigger salinity swings differ slightly, but this rests mostly on Willapa Bay.

< 1%

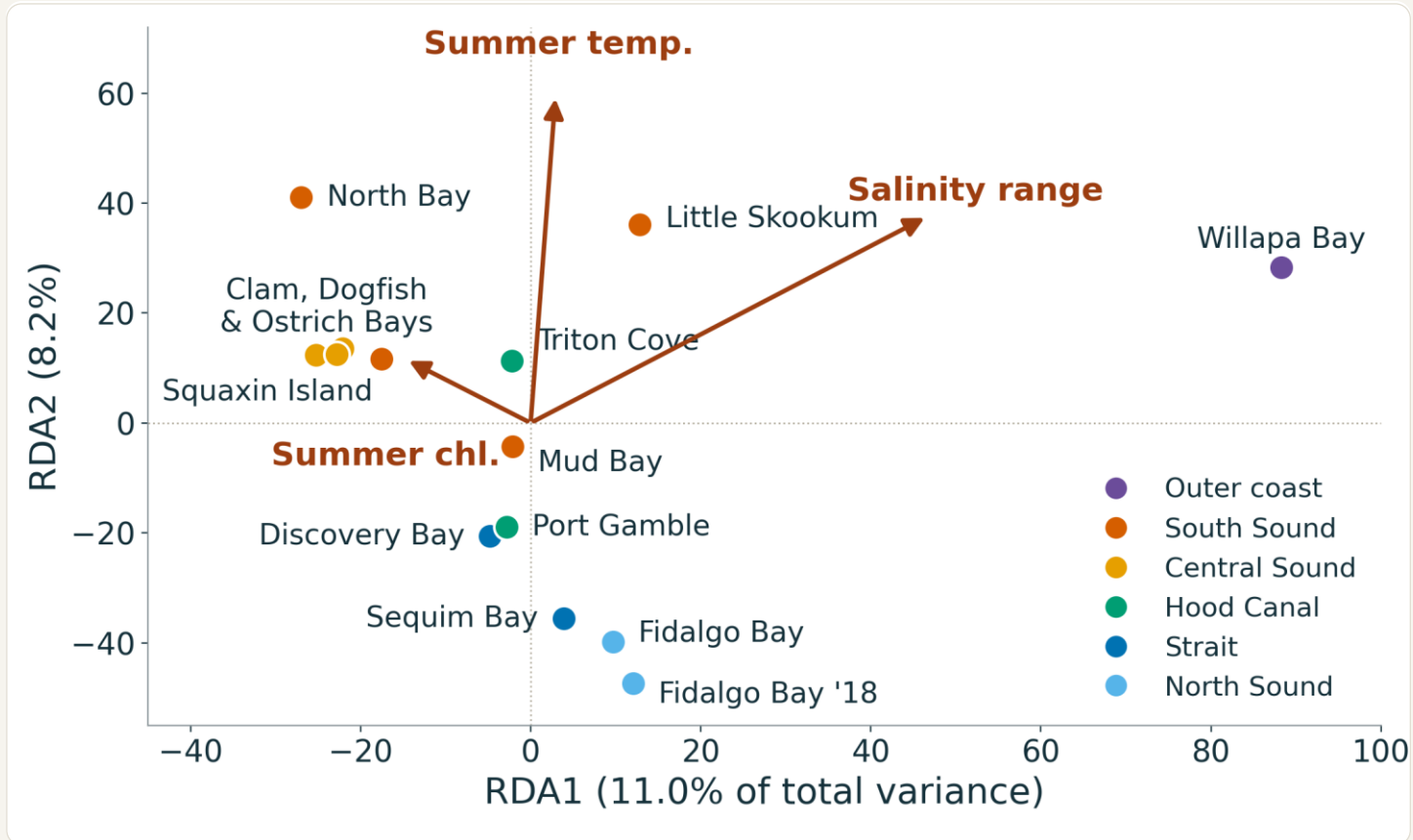
temperature hint

Summer temperature explains under 1% once genetic group is accounted for.

Counting each oyster as independent made the environment look important; treating each bay as one sample removed most of that signal.

REDUNDANCY ANALYSIS

Salinity range carries the only environmental signal



p = 0.024

salinity range, marginal test

11.0%

RDA1, driven mostly by Willapa Bay

p = 0.16

global environment model, 14 sites

READING THE RESULTS

What the results do and don't tell us

What they support

- Genetic groups are real and consistent
- Coast and Puget Sound stocks are distinct
- Water conditions are not a strong guide to genetic makeup

What they don't show

- That oysters are not adapted to local water. Effects may be small or need more sites
- Which sites hold the most genetic diversity.
Differences track collection year (2018 vs. 2024)
- Why Mud Bay groups with Hood Canal

THE TAKE-HOME

Across Puget Sound, shared history, not today's water, explains most genomic differentiation.

At the scale of 14 sites and 2015–2018 surface climatologies, with or without collection year in the model.

CONCLUSIONS

What this means for restoration, and what comes next

For restoration

- Keep outer-coast and Salish Sea stocks distinct
- Use Central–South and Hood Canal–Strait groups as a first frame for seed transfer
- Any environmental adaptation is subtle; don't rely on it alone to match sources

Next steps

- Model collection batch (year, library duplication) explicitly
- Add sites and in-situ loggers to break shared-station confounding
- Trace the Mud Bay–Hood Canal link through transfer records
- More oysters per site for polygenic GEA

Thank you · WDFW · Code and results: github.com/zbengt/oly-lc-WGS

Bengtsson, Martinelli & Roberts · Olympia oyster lc-WGS